

Applied NLP

DATA 641

Lecture 1 — NLP in the real world

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

What is Natural Language Processing?

Natural language processing (NLP) is an interdisciplinary subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and human language, in particular how to program computers to process and analyze large amounts of natural language data.

Wikipedia

NLP tasks

There is a collection of fundamental tasks that appear frequently across various NLP projects.



Text
Classification



Information
Extraction



Conversational
Agent



Information
Retrieval



Question
Answering Systems



Spam
Classification



Calendar Event
Extraction



Personal
Assistants



Search
Engines

JEOPARDY!

Jeopardy!



Social Media
Analysis



Retail Catalog
Extraction



Health Records
Analysis



Financial
Analysis



Legal Entity
Extraction

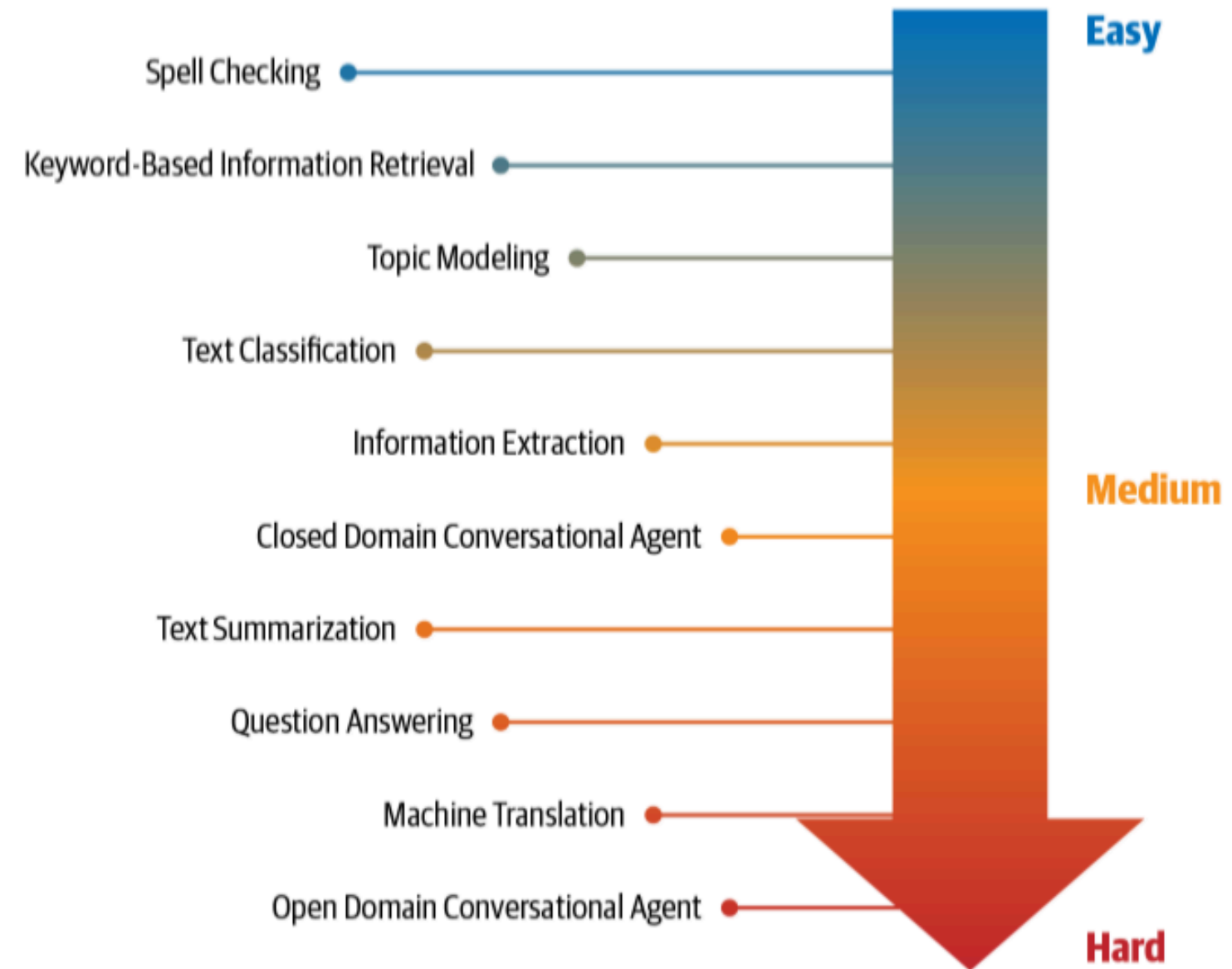
NLP tasks with definitions

- **Language modeling:** This is the task of predicting what the next word in a sentence will be based on the history of previous words.
- **Text classification:** This is the task of bucketing the text into a known set of categories based on its content.
- **Information extraction:** As the name indicates, this is the task of extracting relevant information from text, such as calendar events from emails or the names of people mentioned in a social media post.
- **Information retrieval:** This is the task of finding documents relevant to a user query from a large collection.
- **Conversational agent:** This is the task of building dialogue systems that can converse in human languages.

NLP tasks with definitions

- **Text summarization:** This task aims to create short summaries of longer documents while retaining the core content and preserving the overall meaning of the text.
- **Question answering:** This is the task of building a system that can automatically answer questions posed in natural language.
- **Machine translation:** This is the task of converting a piece of text from one language to another. Tools like Google Translate are common applications of this task.
- **Topic modeling:** This is the task of uncovering the topical structure of a large collection of documents.

Difficulty of NLP tasks



Why Is NLP Challenging?

What makes NLP a challenging problem domain?

Ambiguity: Ambiguity means uncertainty of meaning. Most human languages are inherently ambiguous.

The man couldn't lift his son because he was so **weak**. ———○ Who was weak?

The man couldn't lift his son because he was so **heavy**. ———○ Who was heavy?

Mary and Sue are **sisters**.
Mary and Sue are **mothers**. } ———○ How are Mary and Sue related?

Joan made sure to thank Susan for all the help she had **received**. ———○ Who had received help?

Joan made sure to thank Susan for all the help she had **given**. ———○ Who had given help?

John **promised** Bill to leave, so an hour later he left.
John **ordered** Bill to leave, so an hour later he left. } ———○ Who left an hour later?

Why Is NLP Challenging?

Common knowledge: A key aspect of any human language is "common knowledge." It is the set of all facts that most humans are aware of. In any conversation, it is assumed that these facts are known, hence they're not explicitly mentioned, but they do have a bearing on the meaning of the sentence.

Creativity: Language is not just rule driven; there is also a creative aspect to it. Various styles, dialects, genres, and variations are used in any language.

Diversity across languages: For most languages in the world, there is no direct mapping between the vocabularies of any two languages. This makes porting an NLP solution from one language to another challenge.